SENSIFY LAB

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Wisdom of the LLM Crowd

A Large-Scale Benchmark of Multi-Label U.S. Election-Related Harmful Social Media Content



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IC2S2 July 30, 2026

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Why this matters

1.1B+

posts published daily

PUBLIC OPINION

misinformation can shape voter beliefs

DATASET GAP

limited standardized labels for election misinformation

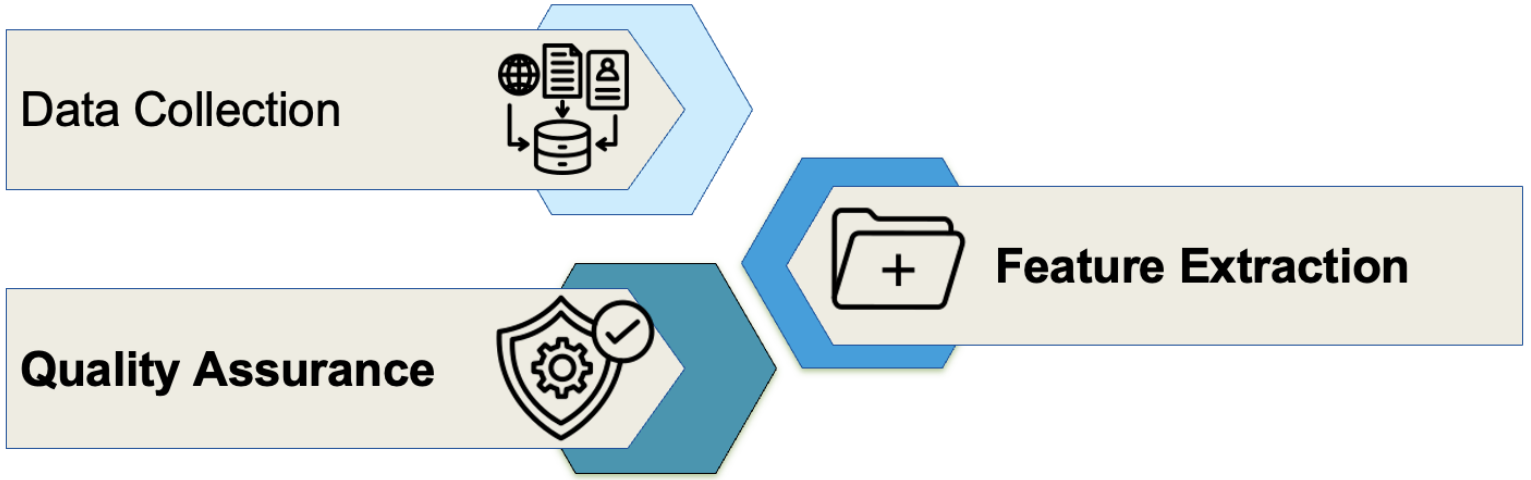
80% of valuable time finding, cleaning, and organizing data (Rezig et al., 2019)

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Our Goal

Develop supportive data annotation pipeline

- Scalable
- Accurate
- Affordable
- Effective



Key Steps in Data Curation

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Case Study - USE24-XD Dataset

Collected nearly **100k** posts from X/Twitter API

During 2024 U.S. presidential election cycle

Focus on harmful political content, Multi-labels

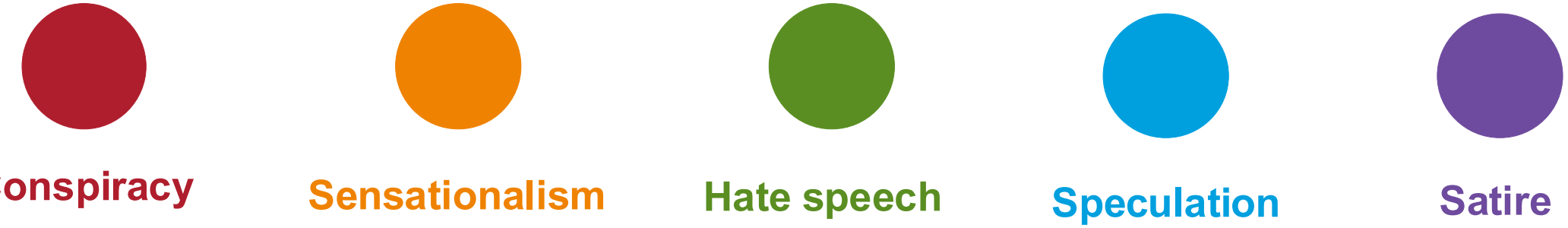
Oct. 17, 2024
collection begins

Nov. 5, 2024
election day

Jul. 16, 2025
collection ends

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Five labels make ambiguity visible



Labels can overlap: a post may receive none, one, or several.

For Example



Subjective Decision

- Conspiracy or
- Sensationalism or
- Hate speech or
- Speculation or
- Satire or

Research Question

- Can *LLMs annotate* online harmful content? Do different LLMs agree?
- How do *human annotators perform* and to what extent do human annotators' *demographics* influence?



LLM Prompt

Your task is to accurately classify social media posts related to the U.S. Presidential Election. Determine whether the given post falls into one or more of the following categories: Conspiracy, Sensationalism, Hate Speech, Speculation, and Satire. Use the detailed definitions provided for each category and respond with True or False for each category only.

Conspiracy: Simplifies complex events by attributing them to secret plots, rejects mainstream information, forms closed belief communities, replaces science with alternative explanations, or frames events as elite deception.

Sensationalism: Uses exaggerated or dramatic language, shock and fear appeal, oversimplifies issues, or employs clickbait-style framing to increase engagement.

Hate Speech: Contains incitement of discrimination, defamation, hostility, or violence based on identity, or makes false statements that damage a person's reputation (libel/slander).

Speculation: Circulates unverified claims for political advantage, driven by partisan interests, amplified in ideological echo chambers, and sustains political controversy.

Satire: Uses humor, political satire, and internet memes to criticize or comment on politics, often spreading through viral online platforms.

Post: {post}

Six LLM models receive the same task

Zero-shot prompting, same label definitions

Temperature = 0

Five Boolean judgments per post



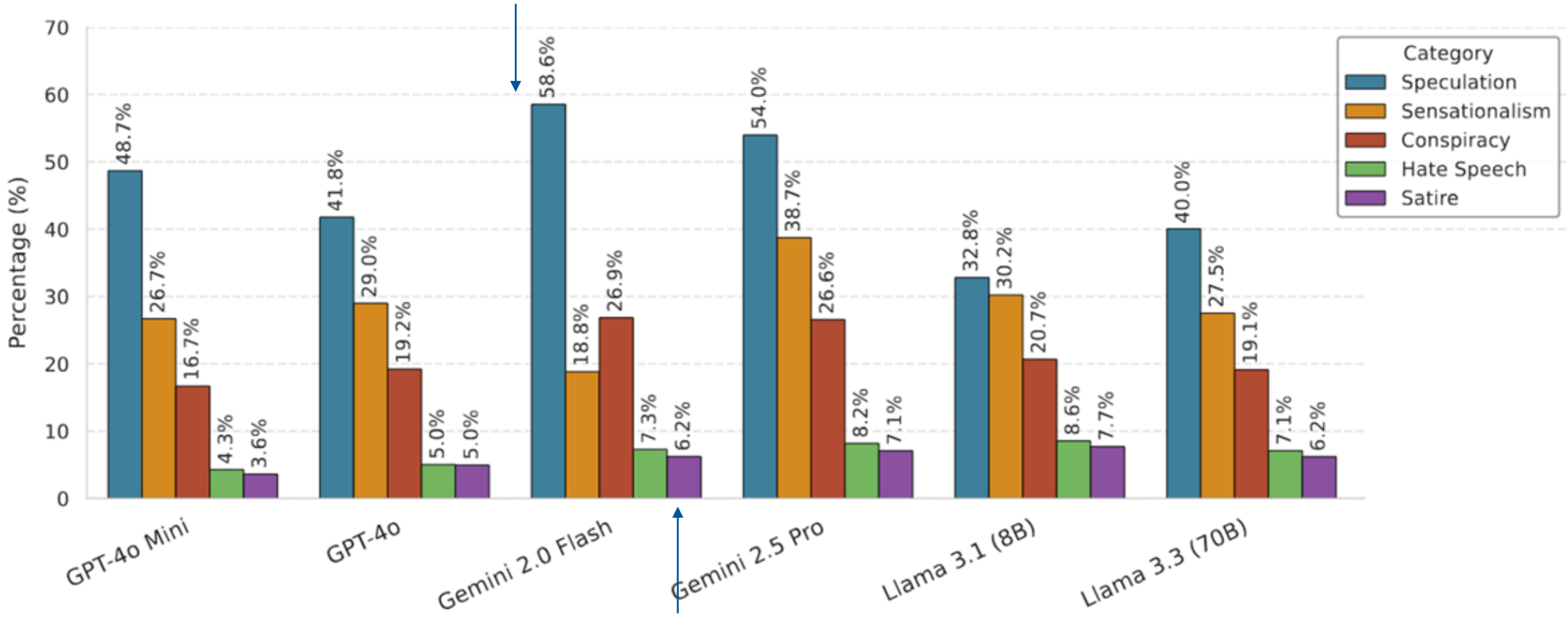
GPT
4o, 4o-mini

Gemini
2.5 Pro, 2.0 Flash

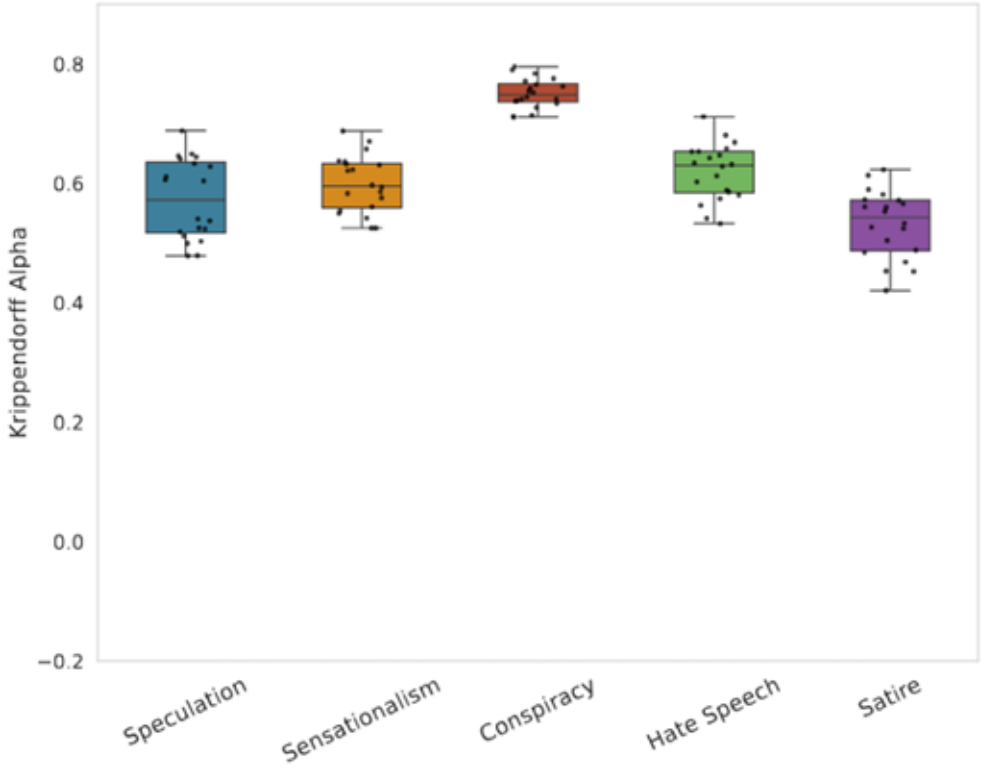
Llama
3.3,(*70 B*) 3.1 (8B)

- Conspiracy ✅ or ❌
- Sensationalism ✅ or ❌
- Hate speech ✅ or ❌
- Speculation ✅ or ❌
- Satire ✅ or ❌

LLM Annotation Distribution



LLM Inter-rater Agreement



(a) LLMs (n=20)

Krippendorff's Alpha (α) - from 3 annotators

Human validation is a reference

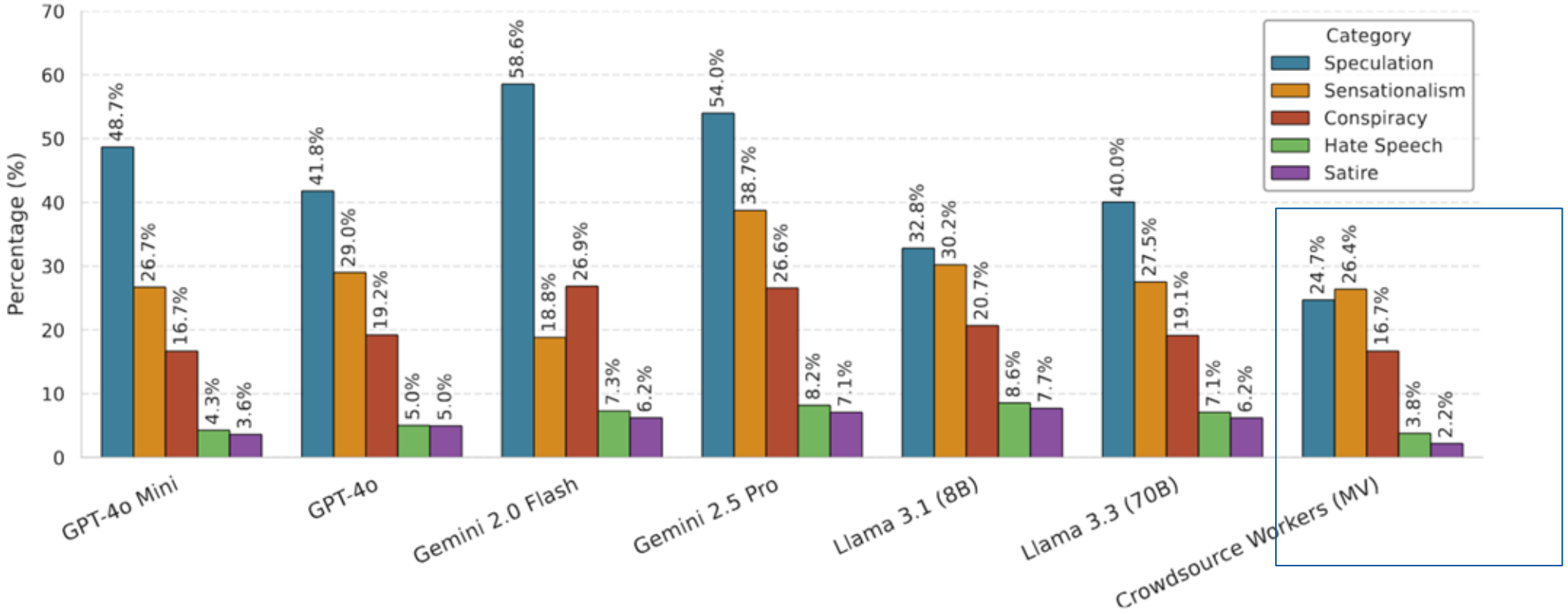
34
qualified Crowdsource U.S. worker

1,000
random posts

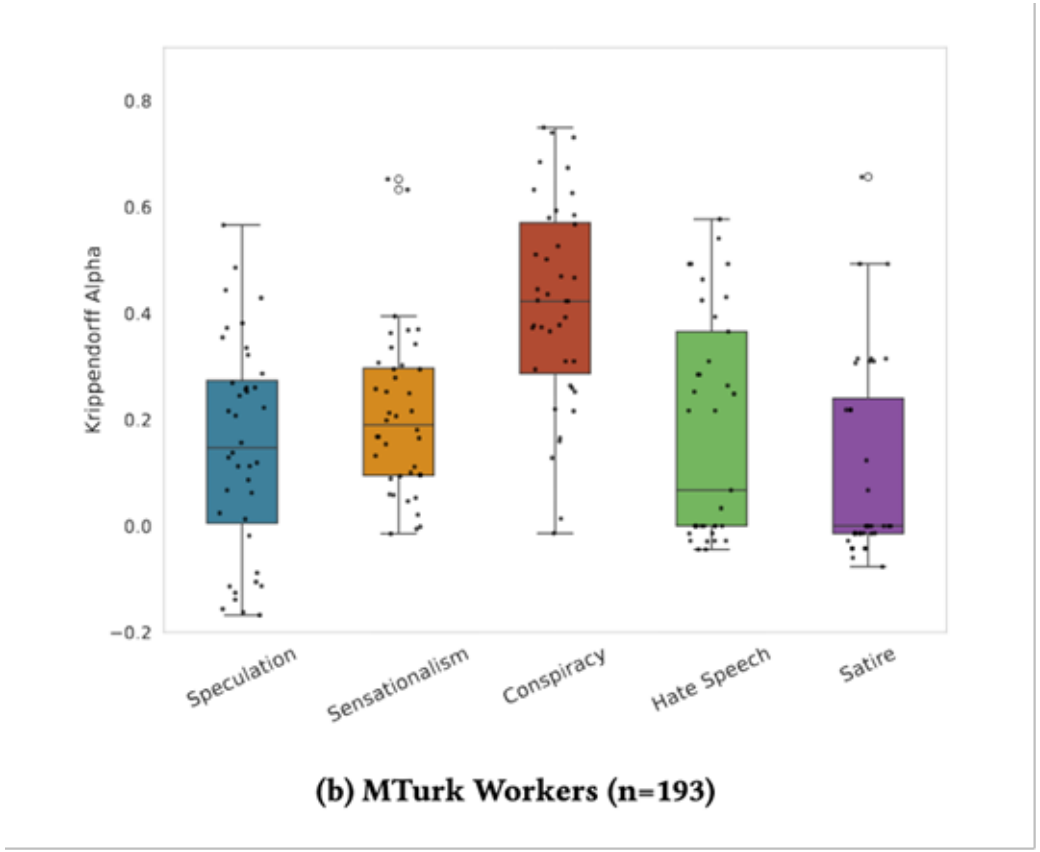
3
judgments per post

The human majority vote is a benchmark, and its variation is evidence.

Distribution for human annotation is similar

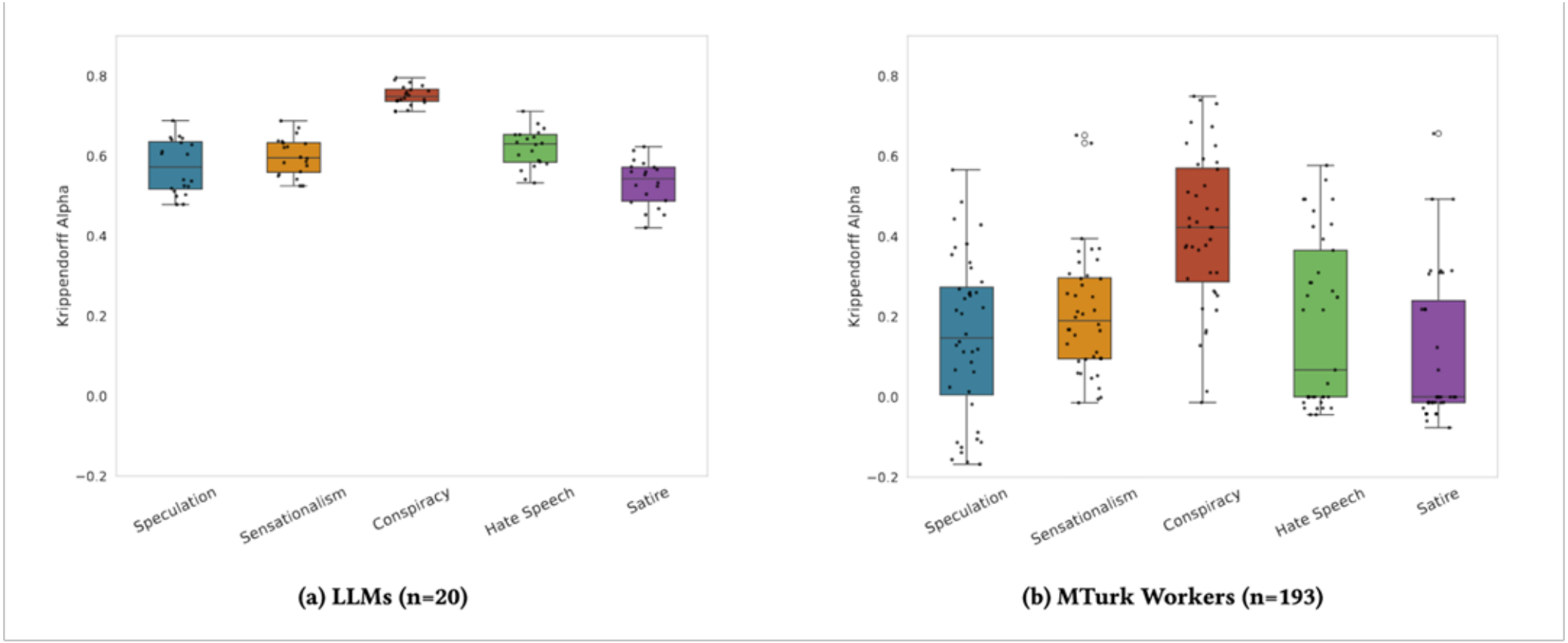


Human Inter-rater Agreement



Krippendorff's Alpha (α) - from 3 annotators

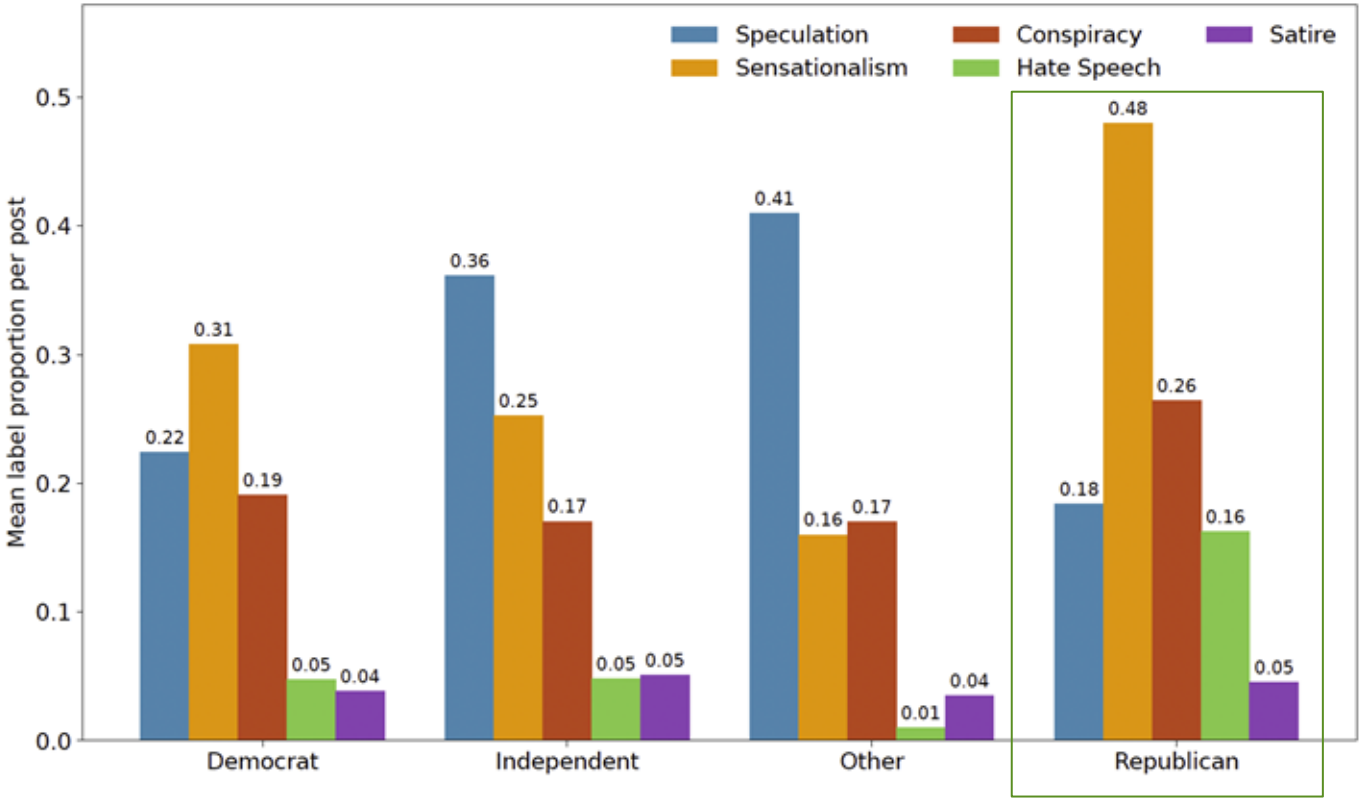
LLMs are more consistent, less discrepancy



Krippendorff's Alpha (α) - from 3 annotators

Demographic Variation

- Most participants are
  - Male (70%)
  - Higher Education
  - 35-44 years
  - Balanced political view
- **Republican assigned more labels in *Sensationalism, Conspiracy, Hate Speech***
- Low to moderate associations for others (Non-significant)



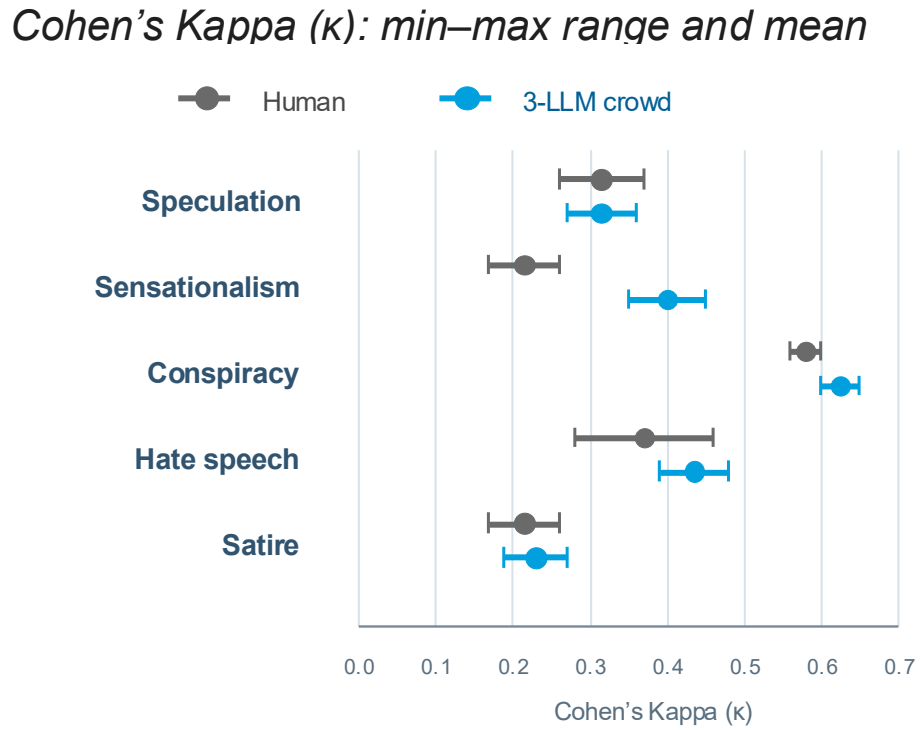
Mean label proportions by Political Affiliation

What about LLM vs Human Agreement?

Human consensus label

Range plot:
- line = min–max
- dot = mean

Result:
- 3-LLM crowd is higher in 4 of 5 categories
- Speculation is nearly identical

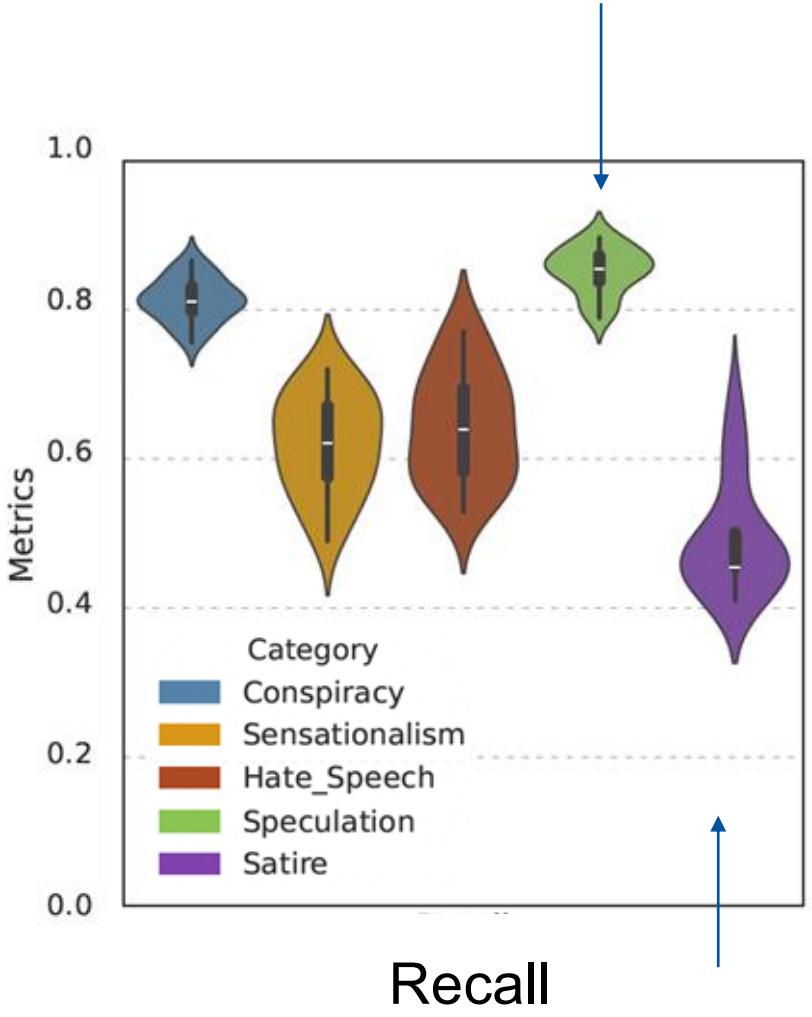


Wisdom of the LLM Crowd

- Compare w/ **majority-vote labels from group of 3 LLMs** (n = 20 groups)

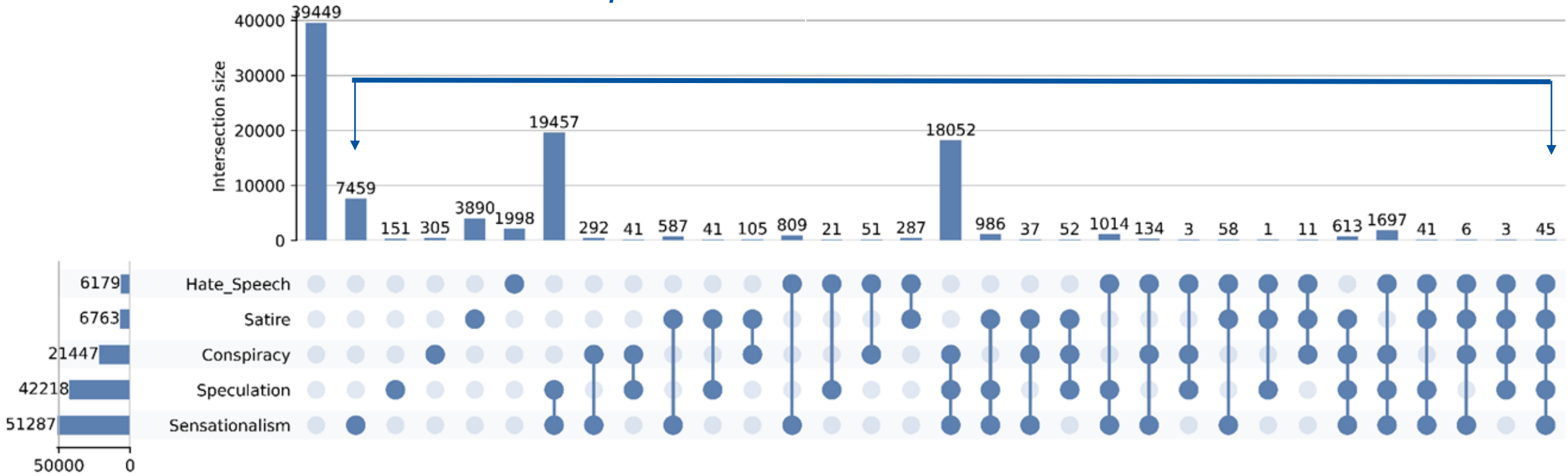
Recall:

- Representative metric - **detection typically focus on the positive class**
- **Reach (up to 0.9) for Speculation**, follow by Conspiracy (0.87)
- Satire (0.68) – most subjective decision



Final Dataset Distribution

62% *of posts receive at least one harmful content label*



Cost–benefit Comparison

Model	Open Source	Parameters (n)	Knowledge Cut-off	Computation Time (Avg)	Cost (Avg)
GPT-4o Mini	×	8 B*	Oct 2023	100 sec	$0.09
GPT-4o	×	200 B*	Oct 2023	88 sec	$1.43
Gemini 2.0 Flash	×	20 B*	Aug 2024	60 sec	$0.05
Gemini 2.5 Pro	×	140 B*	Jan 2025	30 min	$8.97
Llama 3.1	✓	8 B	Dec 2023	130 sec	$0.17 (4× Nvidia L4)
Llama 3.3	✓	70 B	Dec 2023	196 sec	$0.97 (4× Nvidia A100)
Single Crowdsource Worker				7.78 hr*	$130*

- Based on 1000 post
- LLM 318x times Faster, 91x Cheaper
- It would cost $13,000 and 1 month for one human annotation on full dataset

Takeaways



Prompting

Definitions make the measurement contestable before it becomes scalable.



Verification

Combine model families.
1% manual inspection to estimate accuracy



Dataset

USE24-XD
Released